Smart Security System for Vehicles using Internet of Things (IoT)

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***Abstract*—With an ascending advancement of smart technologies, the future of vehicle security systems (VSS) are changing into smart systems for various benefits. With this continuous advancement, internet has become an integral part of one's life where Internet of Things (IoT) is the latest and emerging internet technology that has changed the way one looks at things. Internet of things is developing everyday from small scale machines to large scale machines that can share data and accomplish tasks while individuals are occupied with other activities. The main aim of the paper is to design a smart vehicle security system using IoT, that is to turn a customary vehicle security systems (CVSS) to a smart vehicle security systems (SVSS) for accessing and controlling vehicles remotely using a Smart phone. SVSS are also called as Intelligent Anti-Theft Tracking Systems (iATTS). To be specific, we aim to design a light-weight, low cost, extensible, flexible wireless smart vehicle security system using IoT which employs the integration of Radio Frequency Identification (RFID), Global Positioning System (GPS), Global System for Mobile communication (GSM), wireless communication, cloud networking, and fuzzy algorithm that is used for decision tree. This smart system is created to provide vehicle information such as position, time, and alarm informed to the owner of the vehicle by either using Short Message Service (SMS) or using mobile application. The combination of the above technologies can be used synergistically as a smart vehicle security key to control a vehicle (lock or unlock one's vehicle with the help of SMS/app) from remote locations. The complete system is designed considering all types of vehicles by providing a simple, effective ease of installation, to provide vehicles extreme security and SVSS will be a means for preventing, detecting and counter-measuring robbery of vehicles.**

***Index Terms*—Smart vehicle security systems; Automation; Internet of Things; IoT; Radio Frequency Identification; RFID; Global Positioning System; GPS; Global System for Mobile communication; GSM; Short Message Service; SMS; cloud networking; fuzzy algorithm.**

I. Introduction

Widespread theft of vehicles is a proof that the security arrangement of vehicles, particularly two-wheeled vehicles is as yet deficient. The requirement for the extra security framework is viewed as extremely important to maintain a strategic distance from burglary of vehicles. Such conditions require vehicle owners to give careful consideration to vehicle security. The current scenario needs a good solid security framework for all types of vehicles.

Recent survey says, India is the largest two wheeler market in the world, with sales close to 17.7 million two wheelers in the year 2016, with 48,000 units sold every day. China is the second largest market for two wheelers. Followed by Pakistan, Vietnam, Indonesia, Taiwan and other Asian countries. With more number of bikes on road, comes the risk of theft. According to the report published by the National Crime Records Bureau (NCRB), in the year 2016 alone 211,844 two wheeler vehicles were stolen in India. Out of which only 46,436 vehicles were recovered, which means 78% of the vehicles were unrecovered (this is on the rise when compared to reports from 2010 to 2015).

Typically, two-wheelers are stolen right off streets or apartment parking lots. Vehicles are also being stolen by a towing service or wheel clamping (or vehicle clamping). By the time the police are alerted (which could be a few hours since the theft), the vehicles are either dismantled or sold at throw-away prices, leaving almost no traces. Thus making it difficult to recover the vehicles, leaving the owner and police helpless in bringing the thief to book. In the worst case, vehicles are being stolen by a towing service (authorized by a police officer) and the vehicle owners are not notified.

Interestingly, vehicle theft cases reported all over the world are increasing at an alarming rate and the story remains same for rest of the world. Most of the vehicle security systems are best suit for four wheelers. Many security systems have been proposed over the years, but almost all the recent advanced security systems are designed especially for cars. There are serious limitations with the current security systems for two wheelers.

A smart vehicle security system is a means that enable individuals to control vehicles smartly and automatically within a global or local environment. Many existing, well-established vehicle security system frameworks depend on various technologies but lack security and does not support individuals completely. Reacting to the depiction of the vehicle thefts, numerous security tools have been developed in vehicles, regardless of whether they are simple or they utilize any recent technologies. Developers have already given customary security systems to vehicles, for example, keys to the two-wheel vehicles, and alert systems for four-wheel vehicles. However, security systems [1] provided by the developers are less assuring in this modern technological era; there should be additional security if it is to truly secure. Nasir and Mansor [2] developed an automatic key for the two-wheeled vehicle by using a microcontroller. Tang et al. [3] developed security systems for four-wheeled vehicles

using wireless sensor network technology, and Sehgal et al. [4] succeeded in developing a security system using SMS to track missing two-wheel vehicles.

GSM is a digital mobile telephony system that is widely used in Europe and other parts of the world. GSM uses a variation of time division multiple access (TDMA) and is the most widely used of the three digital wireless telephony technologies (TDMA, GSM, and CDMA). GSM digitizes and compresses data, then sends it down a channel with two other streams of user data, each in its own time slot. It operates at either the 900 MHz or 1800 MHz frequency band.

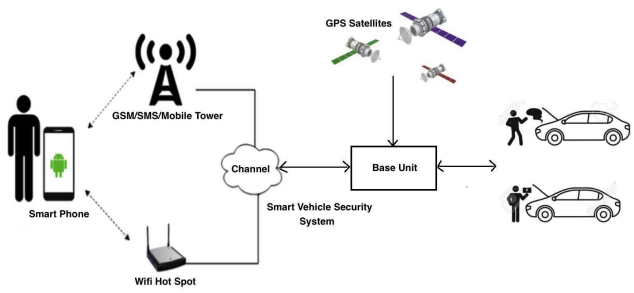

Fig. 1. The concept of the Wireless Vehicle Security System (WVSS) using IoT.

In many developed products, Very Important Person (VIP) security model consists of several layers, that is the first secure point, the second secure point and so forth. This study uses two layers of security for vehicles safety: (i) Authentication method for the first secure point and (ii) Point Positioning for the second secure point. The authentication method has been widely applied to many computer security systems, while the Point Positioning method is one of the methods used in GPS. [5] applied authentication method for wireless network security system and the researchers used Point Positioning method to compare the accuracy of the data processing of GPS-online from some of the GPS data Processing service. The principle of authentication method is the process to access certain systems such as password [6], fingerprint, the retina of the eye [7], facial recognition [8], ID number (unique number) [9] and others. In these studies, the authentication of the ID number is the first step and by using RFID technology, there will be no duplication of ID numbers, or there will be no same ID number. Point Positioning method [10], also called absolute method, is the basic method in GPS which aims to obtain a position in real-time with good accuracy. The description of the security system is that the first safe point uses a standard security system and by applying the authentication method, the rider should first introduce his ID number to turn on the motor vehicle. While the second security point uses GPS for tracking systems position by applying Point Positioning method.

All systems will be controlled by a single component, namely Microcontroller, and as the control of all systems is Radio Frequency Identification (RFID). By applying the Authentication and Point Positioning methods, the vehicles will be more secure, due to the rider must pass through the authentication of ID number to unlock the handlebar of two-wheel vehicles and turn the engine on and there is only one ID number that will be identified and could unlock the handlebar and turn the engine on. Position tracking system in real-time is also an advantage in this system because using Point Positioning method, the position can be identified in real-time with good accuracy. With the two methods applied in the vehicles, the engine will be turned off automatically if the motorcycles are going to be stolen.

The design of the vehicle security system is by using Arduino as the main controller, the features of short message alerts that provide information in the form of warnings and position of vehicle location is worth noting. The system is expected to be an innovative security system for vehicles to realize interactive security systems using various combination of communication technologies. Building IoTs has progressed essentially over the most recent a long time since it has added another dimension to the advancement of of information and communication technologies. Internet of Things (IoT) is a recent communication paradigm that envisions a near future, in which the objects of regular daily life will be furnished with microcontrollers, handsets for computerized correspondence, and appropriate protocols that will make them ready to communicate with each other and with users, turning into an indispensable piece of the Internet. The IoTs have added another measurement to the advancement of information and communication technologies (ICT).

According to CISCO [11], it is assessed that 50 billion objects/devices would be connected with the Internet by 2020. The concept IoT aims at making the Internet significantly more immersive and pervasive. Moreover, by empowering easy access and interaction to a wide range of devices, for example, home electrical appliances, sensors, CCTVs etc, the IoT will foster the improvement of numerous applications that make utilization of the conceivably tremendous amount of data produced by such system to provide new services to various organisations, individuals, and other service oriented sectors [12].

The concept of the Wireless Vehicle Security System (WVSS) using IoT is shown in the Fig. 1. This is a framework that uses PCs, mobiles and/or smart phones to control vehicles smartly, intelligently and automatically through internet from anyplace around the globe to enhance security systems in vehicles. Among many loT applications, smart security systems play an important role in realizing smart cities across the globe.

Current Problems/Challenges for using Smart Security Systems for two-wheelers:

1) Issues with recent technology: There are many issues with the current security system that involves high-end technology.
   a) using image processing technology (biometric techniques such as finger print detection system along with face detection and voice recognition) - these security systems are complex, costly and cannot be implemented on two wheelers.

b) current vehicle security systems can only immobilize the engine and sound a loud alarm, which is a serious limitation.
c) fuel theft
d) equipments/Hardware parts theft (eg. tyres, mirrors, brakes, lights, helmets, etc.)

2) Expensive: The current security systems are quite expensive. The price of the security system should be reasonably low or else the automobile manufacturers cannot implement such a system, as it will increase the overall cost of the vehicle by a big margin. If the design of the security system is such that it is compatible with most of the brands and classes of vehicles then it helps reducing the Non-recurring engineering (NRE) cost.
3) Power Consumption: The power consumption also plays a pivotal role. The overall power consumption should be less as the source of supply for the security system is the 12V battery of the vehicle.
4) Space Utilization: Two wheeler vehicles offer very less space to install the security module and hence area is one of the major constraints. The demand is to design a system that performs necessary function, simple to operate, reasonably priced and small enough to be placed under the seat of the vehicle. Small size of the module allows it to be placed under the seat of the vehicle, there by needing no physical changes to be done to the vehicle.
5) Vehicle Insurance Companies: With thefts occurring at an alarming rate, there is significant expenses involved for insurance companies.
6) SMS/Data Pack Subscription Rates: Individuals need to pay the subscription on a monthly or yearly basis to get regular theft SMSs and to the mobile app.

Goal of the Project: The only possible way out of this problem is to design and develop a simple, reliable and robust security/theft control system that is low cost and effective for individuals to track their vehicles and lock them completely. Then the information captured by the vehicle owner can be quickly shared with the police officers.

The main aim of the paper is to design a smart vehicle security system using IoT, that is to turn a customary vehicle security systems (CVSS) to a smart vehicle security systems (SVSS) for accessing and controlling vehicles remotely using a Smart phone. To be specific, we aim to design a low cost, extensible, flexible wireless smart vehicle security system using IoT which employs the integration of Radio Frequency Identification (RFID), Global Positioning System (GPS), Global System for Mobile communication (GSM), wireless communication, cloud networking, and fuzzy algorithm that is used for decision tree. This smart system is created to provide vehicle information such as position, time, and alarm informed to the owner of the vehicle by either using Short Message Service (SMS) or using mobile application. The combination of the above technologies can be used synergistically as a smart vehicle security key to control a vehicle (lock or unlock one's vehicle with the help of SMS/app) from remote locations. This complete system is designed considering all types of vehicles by providing a simple, effective ease of installation, to provide vehicles extreme security.

The various technology exposures that the paper provides include, Internet of Things (IoT), Wi-Fi Technology, RF Transceiver, Arduino Open Source Electronics Platform, Android Application Development and Relay Switching Principle. The details are explained in the design and implementation section.

II. Design and Implementation

The general features of the smart security system for vehicles is given in the Figure 2.

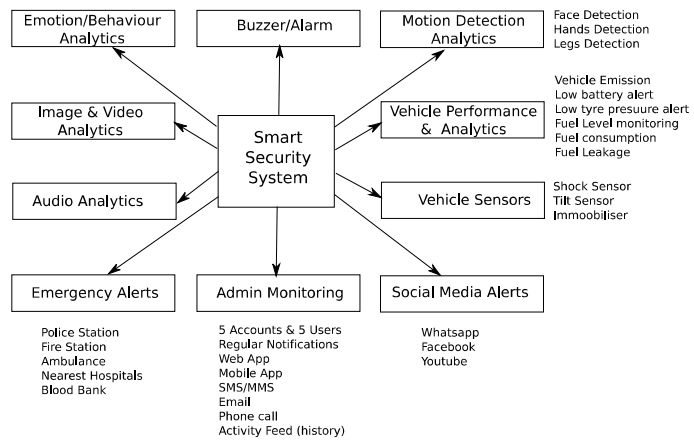

Fig. 2. Features of the Smart Security System for Vehicles using IoT.

The design of double security system based on RFID will be applied to the predetermined vehicle model namely the motor manufactured that has a large enough size of the storage box. Double security system on vehicles based on RFID is an open-loop control system in which the control system refers to the detection output results provided by the RFID detection system, and there is no feedback sent back to the correction process. The Fig. 3 represents the block diagram of an overall smart vehicle security system. An interactive double security system for vehicles based on RFID consists of several subsystems namely identification, detection, controllers, and output. subsystems.

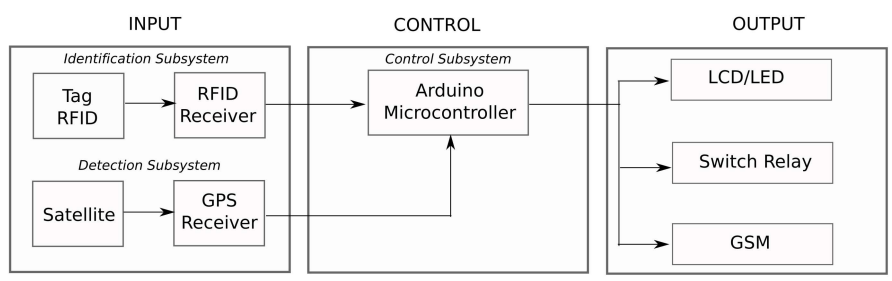

Fig. 3. Block diagram of an overall smart vehicle security system

Identification Subsystem: The identification subsystem is a part of an input subsystem. RFID detection subsystem consists of RFID tags, RFID Reader, and the main controller Microcontroller Arduino Uno R3. RFID Reader will read the RFID tag identification data of RFID tag information to be sent to the Arduino Uno R3 microcontroller as a controller.
Detection Subsystem: GPS detection subsystem consists of satellites, GPS Receiver and the main controller Microcontroller Arduino Uno R3. The satellite will detect the

presence of a GPS receiver and transmit data in National Marine Electronics Association (NMEA) format and Global Positioning System Fix/Accuracy Data (GGA). The data is then sent to Microcontroller Arduino Uno R3.

Output Subsystem: Output subsystem consists of Minimum Output, Relay Action and GSM Communication subsystems. There are LED and Buzzer on Minimum Output Subsystems as indicators which indicates the input and a 16x2 LCD which indicates the data viewer output. By using a relay as automatic switches, action subsystem, a subsystem that serves as a breaker/ignition circuit connecting the vehicles, can be realized.

Control Subsystem: Arduino Uno microcontroller is capable in providing an initial output of 5 volts (high condition) as a trigger to activate the relay so that the motor circuit ignition systems can be connected. The circuit is connected to the vehicle key switch so that the vehicle ignition system will only be activated when the ignition relay and the circumstances are connected. GSM communication subsystem serves as a tool that supports communication between the microcontroller and the mobile phone of the vehicle owner. Communication subsystem is realized by using Linksprite ATWIN GSM Shield module compatible for Arduino Uno R3 microcontroller. GSM module requires ATcommand protocol in the practice of its functions, so it should be described in the program code of Arduino microcontroller.

Also, the security system installed in vehicles can be used to monitor vehicle location using our own mobile application "Find my Vehicle". So, our app will help one to locate the device on a map, remotely lock it, play a sound, display a message and/or get driving directions to vehicle location. The satellite station acquires the location details and can produce it in Google Maps. In this system the Base station and the Satellite station operate in a Master-Slave configuration where the base station provides the necessary commands to satellite stations to acquire locations. The system is controlled by an Android Smartphone which acts as a Client and data are sent via Socket programming. The App provides an interface for each of the three stations using which user has access of controlling and monitoring them remotely (eg. the concept of "find my iphone app"). There is also an another communication module, GSM module, as a backup if there is no internet coverage for updates from the system.

III. Results and Analysis

Tools and systems test was conducted in stages ranging from the test on components/tools, subsystem test, and test of the overall system.

A. Test on Component/Tools Function

Test on the function of components/devices/tools was conducted to avoid errors resulting from non-functioning of one or more of the components/tools. The test was conducted using digital multimeter measurement tools and software for each device.

1. Voltage Source Test: The test was conducted to determine the voltage capacity from voltage source whether it can provide the voltage needed by the system. The voltage sources are connected to the voltage regulator circuit that consists of several components of the voltage regulator IC 7805, 7809, and 7812. This is conducted to adjust the voltage requirements in the system. From the test results of the voltage source, the error values were be obtained using:

$$Error = |IdealVoltage - AverageMeasuredVoltage|$$

2. Tags and RFID Reader Test: This test used three units of RFID tags. The test was carried out by using RFID Tester software. RFID minimum circuit was connected to one unit of the portable computer ACER 4752G research. RFID minimum circuit transmit the data contained on the RFID tag serially to the computer unit via a USB to Serial RS232. In the first RFID tag test data, the data are obtained with the card number 8803460 (tag1). In the second RFID tag test, the data are obtained with the card number 8803459 (tag2). In the third RFID tag test, the data are obtained with the card number 7206252 (TAG3).

3. Microcontroller Arduino Uno R3 Test: Main controller Microcontroller, Arduino Uno R3 Test, aims to determine the conditions of Microcontroller Arduino Uno R3 whether it is in good condition or not. Test main controller Microcontroller Arduino Uno R3 was conducted using software IDE (Integrated Development Environment) 1.0.5 Arduino. Arduino Microcontroller conditions can be determined by plugging it into the computer and upload the sample program to the Arduino Uno R3 with the Arduino IDE software.

B. Subsystem Test

Subsystem test was conducted to ensure that it can perform its function according to the needs of the system. Tests are carried out on detection subsystem, a control subsystem and output subsystem then observing whether the output results are by the system or not.

1. Test on Identification RFID Subsystem: In the subsystems test a unit RFID Reader ID-12 LA, an Arduino Uno R3 microcontroller unit, a series of minimum output systems, and software Arduino IDE 1.0.5 are used in the test. This test was conducted to identify the ability of RFID Reader for reading RFID tags then transmit the ASCII data information to the microcontroller Arduino Uno R3 and displayed on a series of Arduino IDE 1.0.5 software monitor.

2. Test on GPS Detection Subsystem: A unit of U-box GPS Receiver CN-06, an Arduino Uno R3 microcontroller unit, Arduino IDE 1.0.5 software, computers, and software U-Center GPS Evaluation Software are used in the test. Data retrieval GPS (Global Positioning System) in the form of latitude and longitude data is carried out at several points in the area of Bangalore, India (random).

3. Test on Action Relay Subsystem: A 12-volt voltage source, relay circuit Activator, a unit of Microcontroller Arduino Uno R3, computers, and software Arduino IDE 1.0.5 are used in the test. The test was conducted to determine the performance of the circuit as a switch connecting Relay movers and breaker based on the instructions given by the main control of Microcontroller Arduino Uno R3. The command is affected

by the character input received from the computer via serial communication by using a USB. Character "1" makes the relay to be in connecting position so that the LED lights on, while the character "0" makes the relay back to be in disconnecting position so that the LED light is off.

4. Test on GSM Communications Subsystem: The unit Microcontroller Arduino Uno R3, Linksprite ATWIN Quadband GPRS/GSM Shield module, a mobile phone, a computer, Arduino IDE 1.0.5 software and SSCOM32E software were used in the test. The test was conducted to ensure that the Linksprite ATWIN Quadband GPRS/GSM Shield module can realize the communication between the microcontroller Arduino with the mobile phone.

Linksprite ATWIN Quadband GPRS/GSM Shield is a communication module that is compatible with microcontroller Arduino Uno R3. The module supports GSM service. Linksprite Quadband GPRS/GSM Shield supports SMS in Text format and PDU (binary). Linksprite ATWIN Quadband, GPRS/GSM Shield, has a frequency of 900 MHz which is very supportive for use in India.

C. System Test

System test is a test conducted after integrating the existing subsystems into one. The time delay that occurs in the system is counted in this stage. It can be seen that there is a time delay for some of the action given as a form of response to some of the conditions that feed into the system.

Preparation Time delay is a time delay that occurs when a new system is turned on. Time delay relay On is a time delay required for the input conditions of RFID tag is correct, and the key is switched on so that the relay becomes active. Delay time relay Off 1 is the time delay required for the key of the motorcycle is switched off so that relay becomes inactive. Delay time relay Off 2 is the time delay required for the input conditions of GSM character input are met so that the relay becomes inactive. The system encountered a delay time of preparation by an average of 5.6 seconds after getting power from a voltage source. The system can provide action in response to the input conditions that occur with the average time delay of 1.6 seconds.

Microcontroller Arduino Uno R3 provides the action by connecting Relay if the data of RFID tag is appropriate. The next action is to make phone calls to the phone number registered if the RFID tag data is corresponding and the vehicle is On. The other system action is to send information about the location of the vehicle in the form of a short message to the phone numbers listed.

IV. Discussion

Based on the results of the tests that have been carried out on the system's for identification, detection, actions and communication, the results obtained is by the provisions of the system functionality. The system can distinguish RFID tags objects as an access key to vehicle engine from ASCII data stored in the RFID tag and then takes action by connecting a series of vehicle engines, the system can distinguish the character of incoming SMS through Linksprite GSM Shield module as input characters and then take action by replying to SMS providing vehicle position information so that the system can communicate with the mobile devices of the vehicle owner's.

The process of identification, detection, and communications on the systems requires serial communication process with the main data processor Microcontroller Arduino Uno R3. The main data processor Microcontroller Arduino Uno R3 has a pair of the pin of serial communication of data receiver (Rx) and the data transmitter (Tx), in which it is capable only of communicating with the process of receiving data at a time. This causes the needs of the system which has three serial communication process (RFID identification, detection of GPS and GSM communications) cannot be obtained. The needs are met by using two pairs of fake pin serial addition by utilizing the program code libraries "SoftwareSerial" and "AltSoftSerial". Pin serial communication works at the same Baud Rate that is 9600. Fake pin serial has the same working principle as the original serial pin found on the Arduino Uno R3, but the pins work interchangeably in a very short time with the risk of delay time (delay) accumulated.

The test results show that there is a time delay experienced by the system work. The time delay is caused by several factors, including the influence of environmental conditions such as building construction and weather capable of disrupting the state of signals for data communications and the effect of time delay program. The performance of the GPS receiver to receive data from satellites and sends it to the microcontroller Arduino Uno R3 are deeply affected by the construction of buildings that block the transmission of radio waves between the GPS receiver and satellites. It also affects the performance of GSM Shield to receive data in the form of a short message or a call during a loss of signal. The time delay in the program uploaded to a microcontroller Arduino Uno R3 is a major time delay of the value of the time delay in the system.

V. Conclusion & Future Work

Based on the test results and discussion, it can be concluded that a double security system has been realized interactively on vehicles which are capable of distinguishing the RFID tag based on the ASCII data saved to turn the vehicle on and able to provide information on the location of the vehicle. The security system is equipped with an emergency safety feature by utilizing GSM communication to turn off the machine of the vehicle when it is stolen. The accuracy of information is affected by environmental conditions as well as the time delay of the system. The time delay of the entire system can be tolerated for the effectiveness of the system with the average value of the preparation delay time of 5.6 seconds and the average delay time of 1.6 seconds. Error acquisition 4.067% on the source system voltage shows that the voltage source system in in good condition.

The idea of the proposed system is to replace the conventional security systems with the Internet of Things (IoT), where the vehicles are connected to the internet and can

be accessed from anywhere by means of an android smart phone which is capable of connecting to the internet through wireless networks. To improve the work, a systematic evaluation and an effective description of component tests will be conducted. Also, we aim to provide a comparison of existing CVSS with our proposed SVSS in an integrated evaluation. This will supports the overall goal of our work. The relationship between the approach of cloud networking and fuzzy algorithms are introduced exclusively and the future work will rely on them. The designed smart security system can be installed in vehicles to monitor vehicle location using our own mobile application "Find my Vehicle". So, our app will help one to locate the device on a map, remotely lock it, play a sound, display a message and/or get driving directions to vehicle location. The satellite station acquires the location details and can produce it in Google Maps. Also the system can store the spatial/temporal data and the status of the vehicles in the thingspeak cloud platform.The stored data is visualized in the mobile application in the form of graphs. The system helps users to analyze and monitor vehicle location anytime anywhere. Our smart security system using IoT has been experimentally proven to work satisfactorily by connecting vehicles, which can be easily controlled remotely/wirelessly through internet. When we consider fleet management and tracking vehicles, artificial intelligence and blockchain solution will play a key role.

The blockchain is a network, a distributed database of records or public ledger of all transactions/messages/notifications that have been executed and shared among participating parties. Each transaction in the public ledger is verified by consensus of a majority of the participants in the system. Once information is entered or registered, that information can never be erased and it is called as Data Integrity. Authentication is that each transaction is digitally signed using the private key of the participant that made the transaction. In particular, smart system for fleet management is extremely important because it is closely related to the safety of the vehicles managed by the fleet aurthorities. However, the data sent and received of existing smart system is vulnerable to forgery and hacking. To improve these security issues, we need to extend this system based on blockchain, especially when these systems judge particular events around itself by using data sent from sensors and then operates. Also, the system with blockchain provides authentication, data integrity and non-repudiation. It prevents an unauthenticated user from participating in the blockchain network. Using this system as framework the following further enhancements can be achieved [13][14][15][16][17][18].

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